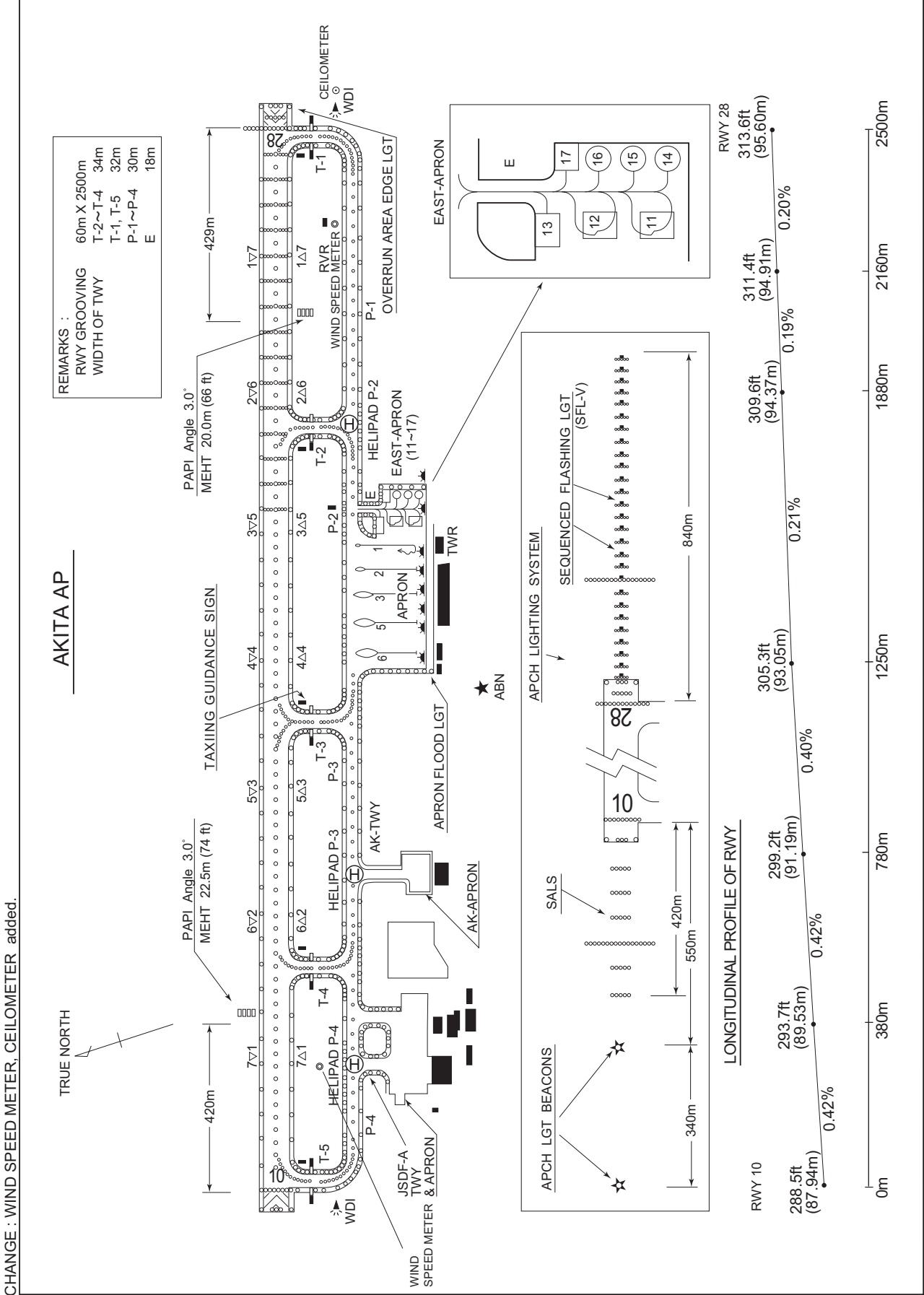


RJSK / AKITA

AD CHART



INTENTIONALLY LEFT BLANK

STANDARD DEPARTURE CHART -INSTRUMENT

RJSK / AKITA

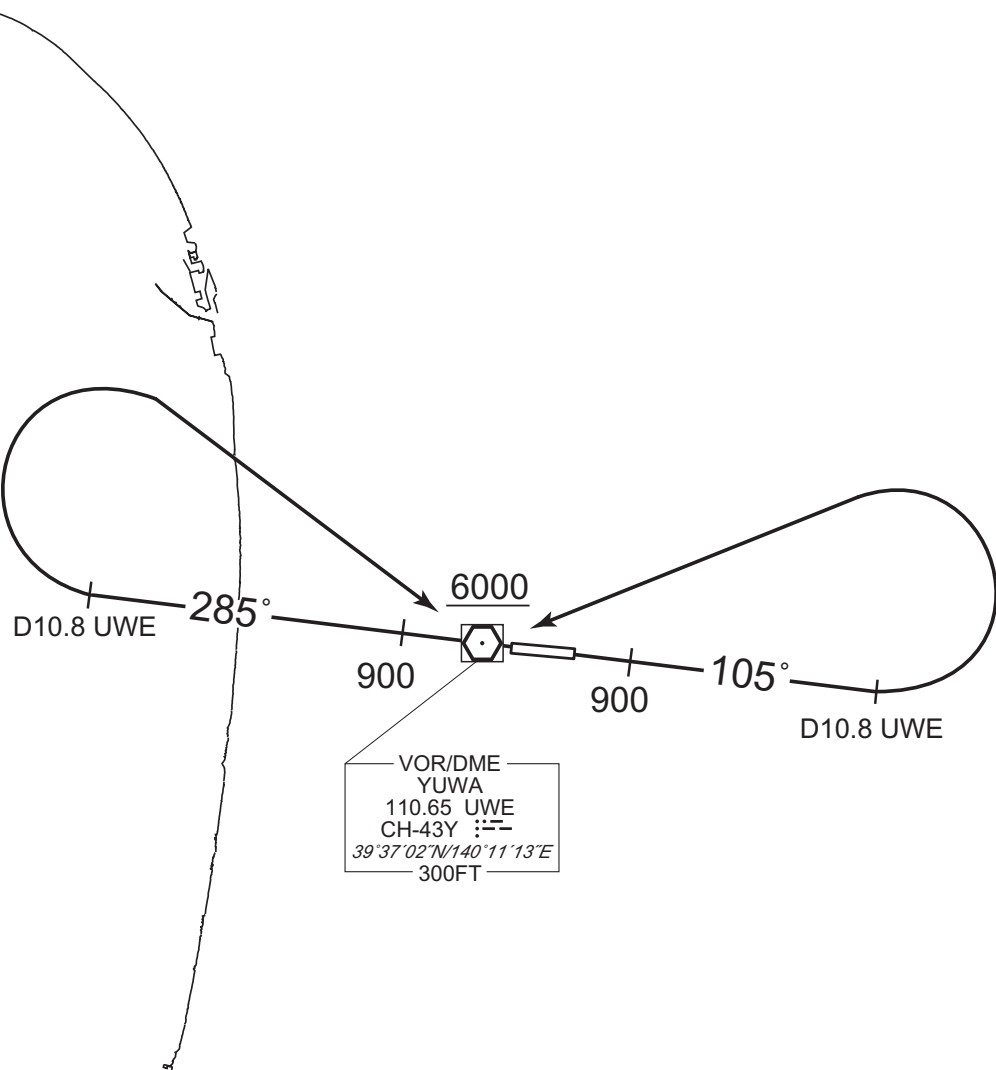
SID

YUWA REVERSAL SEVEN DEPARTURE

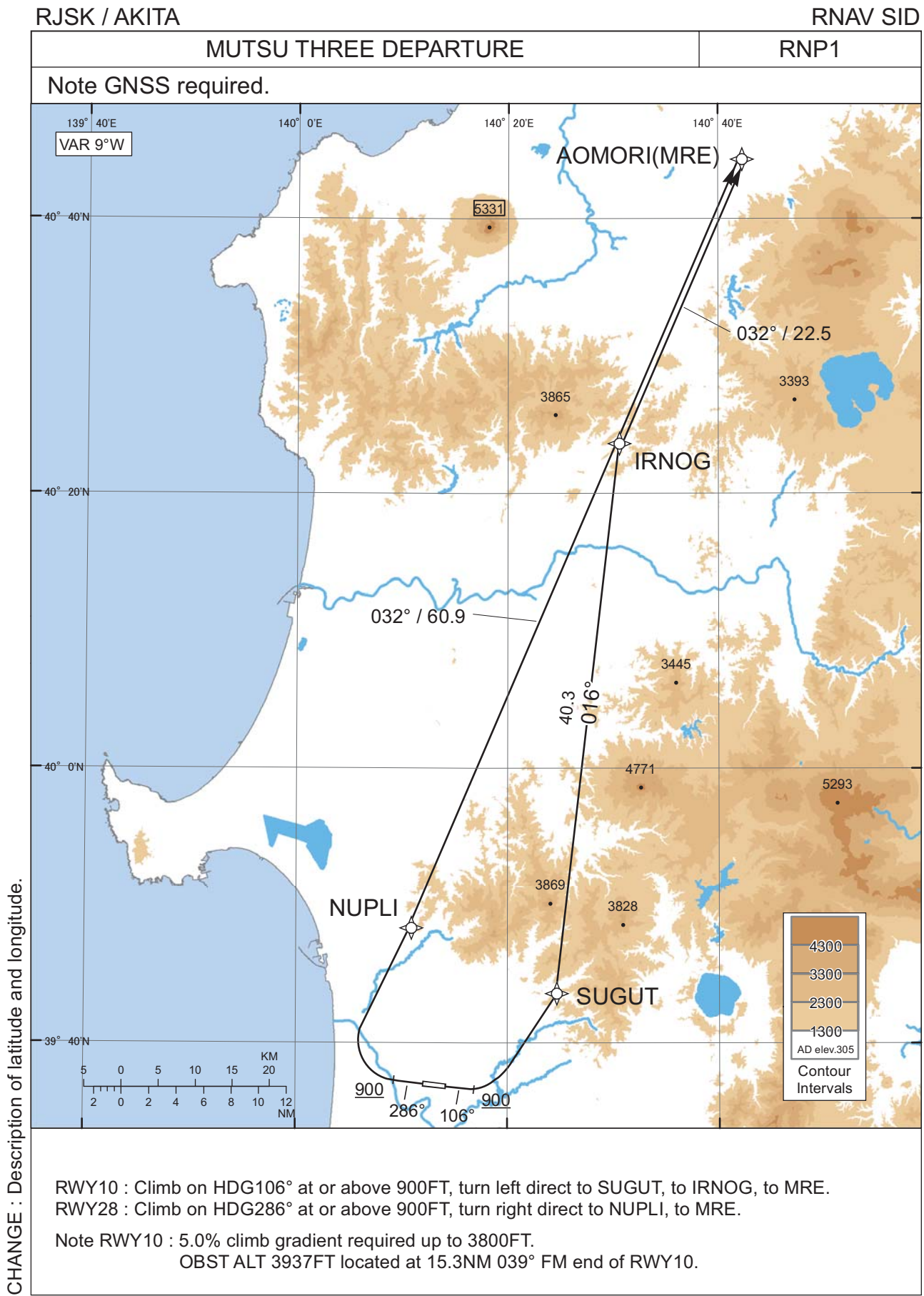
RWY 10 : Climb RWY HDG to 900FT, via UWE R105 to 10.8DME, turn left,...
 RWY 28 : Climb RWY HDG to 900FT, via UWE R285 to 10.8DME turn right,...
 ...direct to UWE VOR/DME.
 Cross UWE VOR/DME at or above 6000FT.

Note RWY10 : 5.0% climb gradient required up to 3300FT.
 OBST ALT 3543FT located at 13.8NM 071° FM end of RWY10.

CHANGE : PROC course. PROC renamed. Note added.



STANDARD DEPARTURE CHART -INSTRUMENT



STANDARD DEPARTURE CHART -INSTRUMENT

RJSK / AKITA

RNAV SID

MUTSU THREE DEPARTURE

RWY10

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	106 (096.6)	-9.1	-	-	+900	-	-	RNP1
002	DF	SUGUT	-	-	-9.1	-	L	-	-	-	RNP1
003	TF	IRNOG	-	016 (006.4)	-9.1	40.3	-	-	-	-	RNP1
004	TF	MRE	-	032 (023.2)	-9.1	22.5	-	-	-	-	RNP1

RWY28

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	286 (276.6)	-9.1	-	-	+900	-	-	RNP1
002	DF	NUPLI	-	-	-9.1	-	R	-	-	-	RNP1
003	TF	MRE	-	032 (023.0)	-9.1	60.9	-	-	-	-	RNP1

Waypoint Coordinates

Waypoint Identifier	Coordinates
SUGUT	394336.3N / 1402441.6E
IRNOG	402338.0N / 1403036.4E
NUPLI	394822.1N / 1401056.5E
MRE	404419.7N / 1404219.2E

CHANGE : Waypoint Coordinates added.

STANDARD DEPARTURE CHART -INSTRUMENT

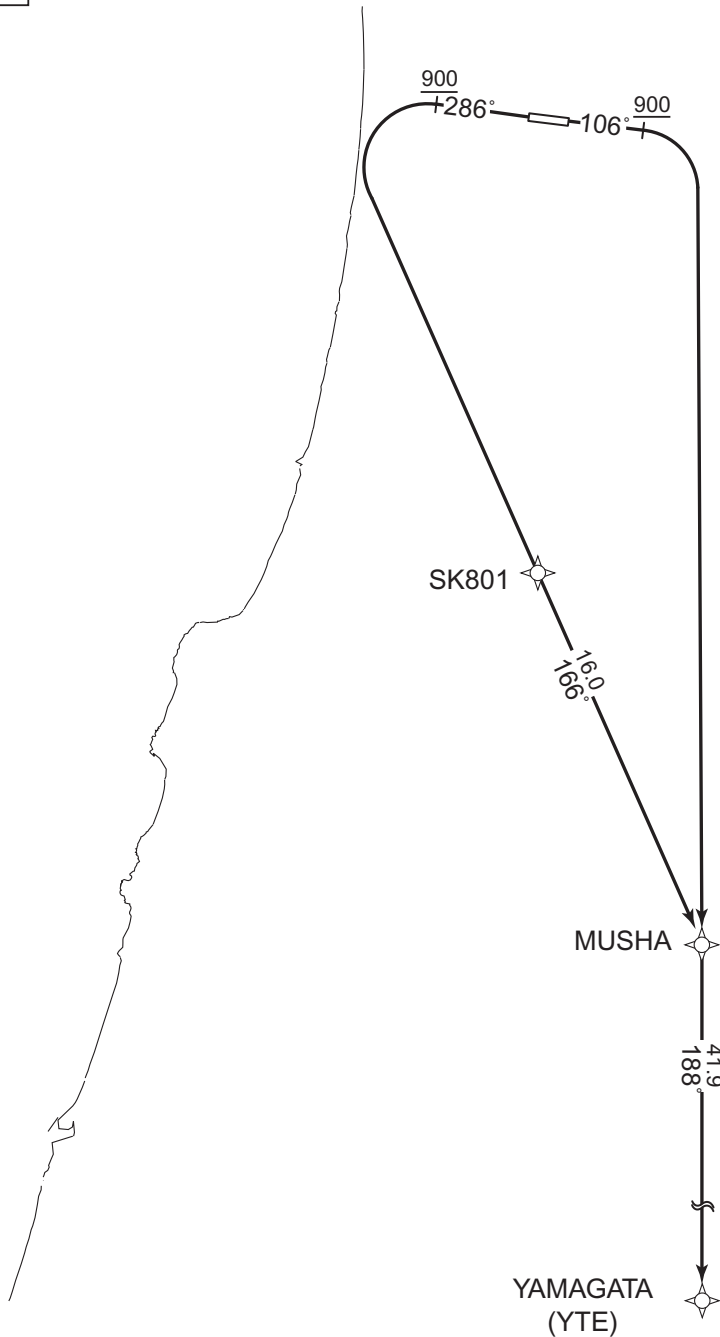
RJSK / AKITA

USYU TWO DEPARTURE

RNAV SID
RNP1

Note GNSS required.

VAR 9°W



RWY10 : Climb on HDG 106° at or above 900FT, turn right direct to MUSHA, to YTE.
RWY28 : Climb on HDG 286° at or above 900FT, turn left direct to SK801, to MUSHA, to YTE.

CHANGE : Description of latitude and longitude.

STANDARD DEPARTURE CHART -INSTRUMENT

RJSK / AKITA

RNAV SID

USYU TWO DEPARTURE

RWY10

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	106 (096.6)	-8.9	-	-	+900	-	-	RNP1
002	DF	MUSHA	-	-	-8.9	-	R	-	-	-	RNP1
003	TF	YTE	-	188 (179.3)	-8.9	41.9	-	-	-	-	RNP1

RWY28

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	286 (276.6)	-8.9	-	-	+900	-	-	RNP1
002	DF	SK801	-	-	-8.9	-	L	-	-	-	RNP1
003	TF	MUSHA	-	166 (157.1)	-8.9	16.0	-	-	-	-	RNP1
004	TF	YTE	-	188 (179.3)	-8.9	41.9	-	-	-	-	RNP1

Waypoint Coordinates

Waypoint Identifier	Coordinates
SK801	391957.0N / 1401250.5E
MUSHA	390511.0N / 1402052.2E
YTE	382319.0N / 1402128.6E

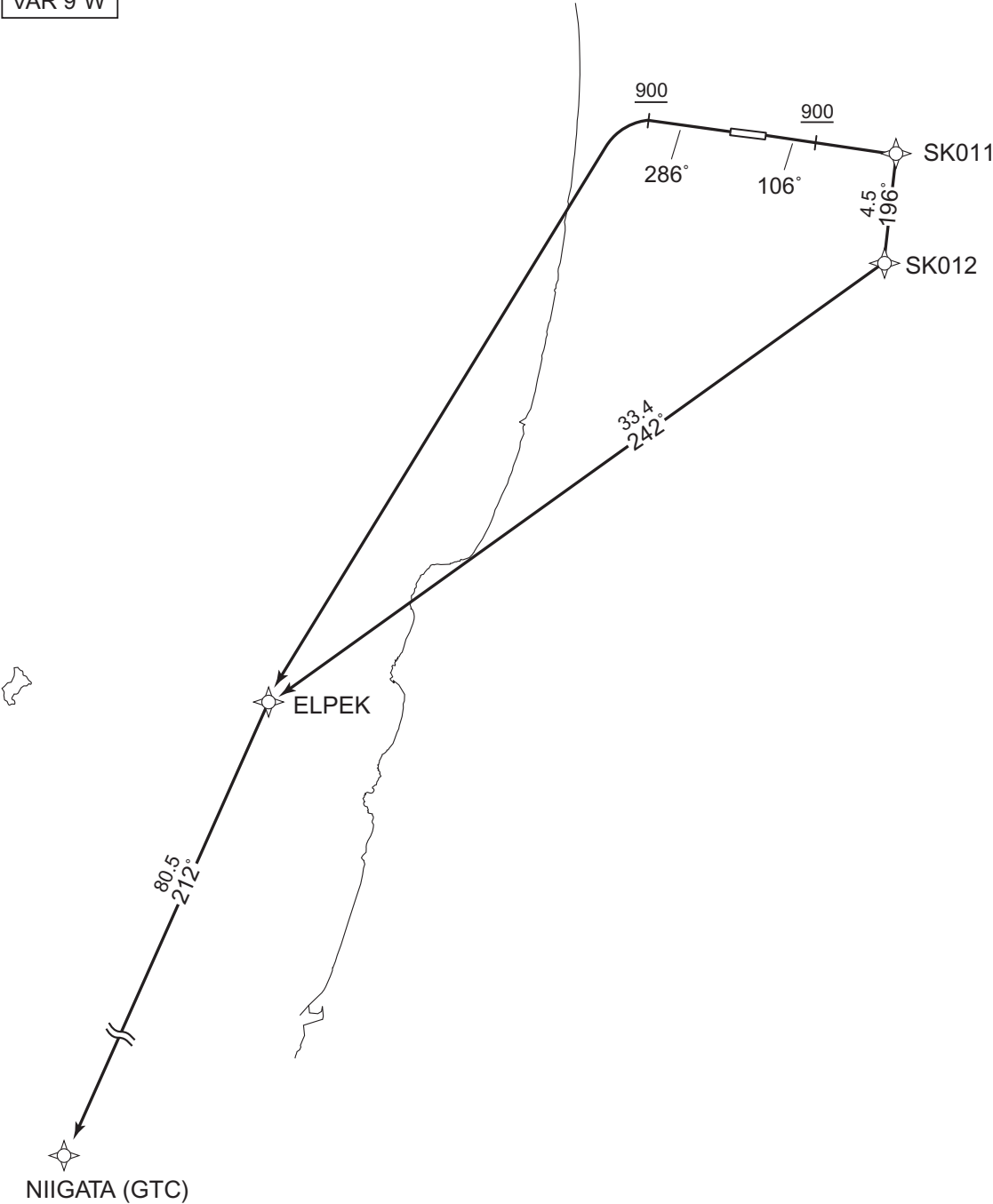
CHANGE : Waypoint Coordinates added.

STANDARD DEPARTURE CHART -INSTRUMENT

RJSK / AKITA	RNAV SID
NIIGATA THREE DEPARTURE	RNP1

Note GNSS required.

VAR 9°W



CHANGE : PROC renamed. ELPEK established. TARAAB abolished.

RWY10 : Climb on HDG 106° at or above 900FT, direct to SK011, to SK012, to ELPEK, to GTC.
RWY28 : Climb on HDG 286° at or above 900FT, turn left direct to ELPEK, to GTC.

STANDARD DEPARTURE CHART -INSTRUMENT

RJSK / AKITA

RNAV SID

NIIGATA THREE DEPARTURE

RWY10

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	106 (096.6)	-8.9	-	-	+900	-	-	RNP1
002	DF	SK011	-	-	-8.9	-	-	-	-	-	RNP1
003	TF	SK012	-	196 (186.7)	-8.9	4.5	-	-	-	-	RNP1
004	TF	ELPEK	-	242 (233.5)	-8.9	33.4	-	-	-	-	RNP1
005	TF	GTC	-	212 (203.0)	-8.9	80.5	-	-	-	-	RNP1

RWY28

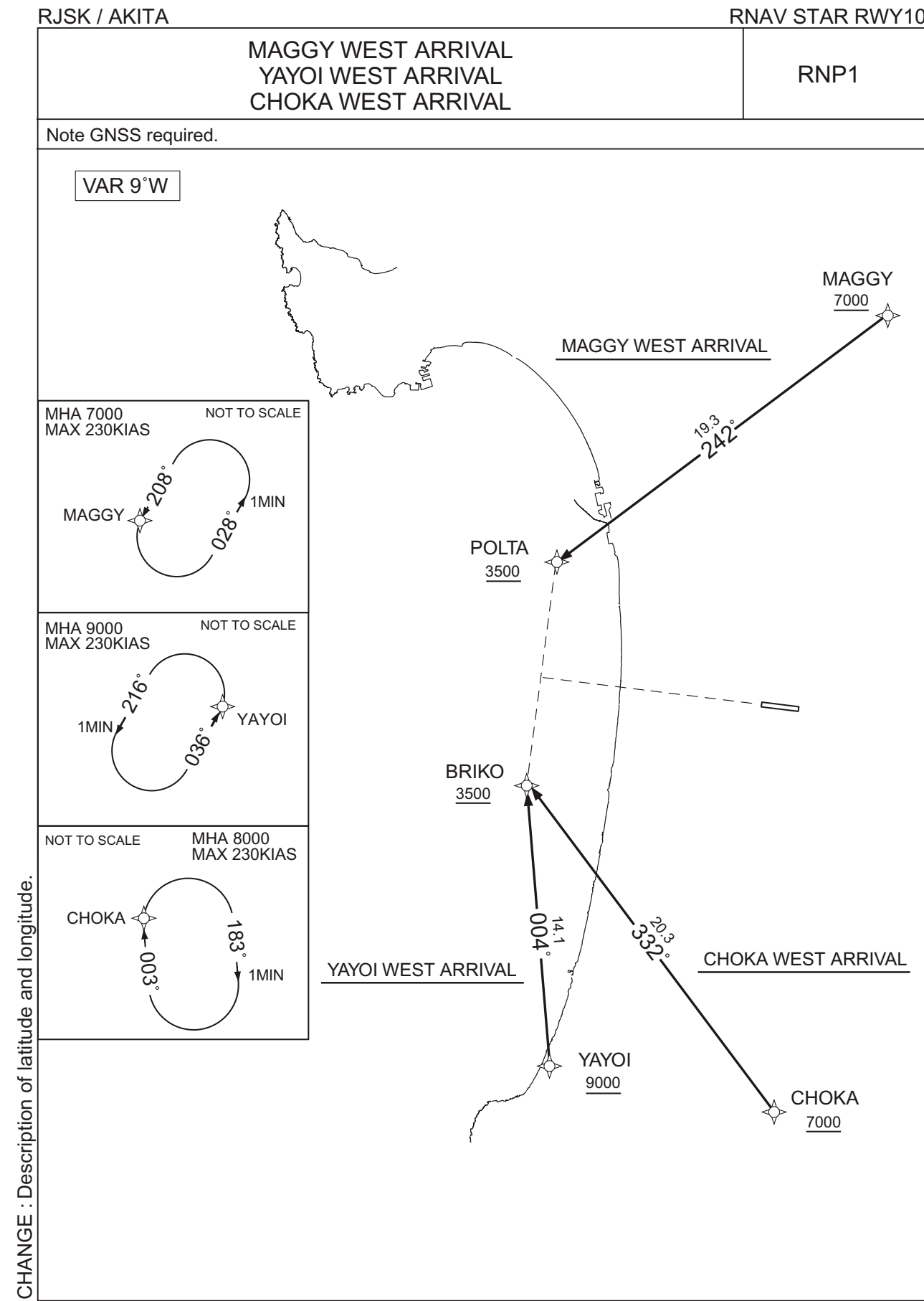
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	286 (276.6)	-8.9	-	-	+900	-	-	RNP1
002	DF	ELPEK	-	-	-8.9	-	-	-	-	-	RNP1
003	TF	GTC	-	212 (203.0)	-8.9	80.5	-	-	-	-	RNP1

Waypoint Coordinates

Waypoint Identifier	Coordinates
SK011	393608.1N / 1402202.4E
SK012	393139.6N / 1402121.7E
ELPEK	391141.2N / 1394642.1E
GTC	375729.9N / 1390653.6E

CHANGE : PROC renamed. ELPEK established. TARA abolished. Waypoint Coordinates added.

STANDARD ARRIVAL CHART-INSTRUMENT



STANDARD ARRIVAL CHART-INSTRUMENT

RJSK / AKITA

RNAV STAR RWY10

MAGGY WEST ARRIVAL

From MAGGY at or above 7000FT, to POLTA at or above 3500FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	MAGGY	-	-	-9.0	-	-	+7000	-	-	RNP1
002	TF	POLTA	-	242 (232.6)	-9.0	19.3	-	+3500	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	MAGGY	208 (199.5)	-9.0	1.0(-14000)	L	7000	FL140	-230(-14000)	RNP1

YAYOI WEST ARRIVAL

From YAYOI at or above 9000FT, to BRIKO at or above 3500FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	YAYOI	-	-	-9.0	-	-	+9000	-	-	RNP1
002	TF	BRIKO	-	004 (355.2)	-9.0	14.1	-	+3500	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	YAYOI	036 (026.6)	-9.0	1.0(-14000)	L	9000	FL140	-230(-14000)	RNP1

CHOKA WEST ARRIVAL

From CHOKA at or above 7000FT, to BRIKO at or above 3500FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	CHOKA	-	-	-9.0	-	-	+7000	-	-	RNP1
002	TF	BRIKO	-	332 (322.5)	-9.0	20.3	-	+3500	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	CHOKA	003 (353.8)	-9.0	1.0(-14000)	R	8000	FL140	-230(-14000)	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD ARRIVAL CHART-INSTRUMENT

RJSK / AKITA

RNAV STAR RWY10

Waypoint Coordinates

Waypoint Identifier	Coordinates
MAGGY	395456.6N / 1401926.8E
POLTA	394311.1N / 1395928.5E
YAYOI	391910.1N / 1395933.8E
CHOKA	391709.5N / 1401401.1E
BRIKO	393314.2N / 1395801.2E

CHANGE : Waypoint Coordinates added.

STANDARD ARRIVAL CHART-INSTRUMENT

RJSK / AKITA

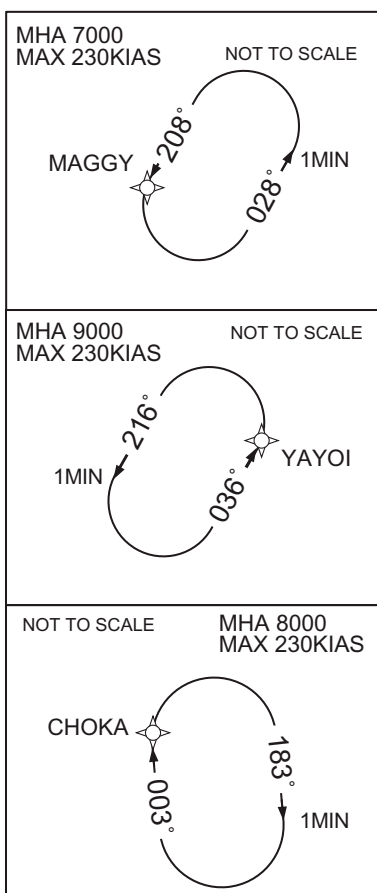
RNAV STAR RWY28

MAGGY EAST ARRIVAL
YAYOI EAST ARRIVAL
CHOKA EAST ARRIVAL

RNP1

Note GNSS required.

VAR 9°W

MAGGY EAST ARRIVALMAGGY
7000

171°

11.9

GAKKO
5000BIJIN
4200YAYOI EAST ARRIVALYAYOI
9000

22.5°

067°

CHOKA EAST ARRIVALCHOKA
7000

15.9

039°

CHANGE : Description of latitude and longitude.

STANDARD ARRIVAL CHART-INSTRUMENT

RJSK / AKITA

RNAV STAR RWY28

MAGGY EAST ARRIVAL

From MAGGY at or above 7000FT, to GAKKO at or above 5000FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	MAGGY	-	-	-9.0	-	-	+7000	-	-	RNP1
002	TF	GAKKO	-	171 (162.1)	-9.0	11.9	-	+5000	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	MAGGY	208 (199.5)	-9.0	1.0(-14000)	L	7000	FL140	-230(-14000)	RNP1

YAYOI EAST ARRIVAL

From YAYOI at or above 9000FT, to BIJIN at or above 4200FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	YAYOI	-	-	-9.0	-	-	+9000	-	-	RNP1
002	TF	BIJIN	-	067 (058.5)	-9.0	22.5	-	+4200	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	YAYOI	036 (026.6)	-9.0	1.0(-14000)	L	9000	FL140	-230(-14000)	RNP1

CHOKA EAST ARRIVAL

From CHOKA at or above 7000FT, to BIJIN at or above 4200FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	CHOKA	-	-	-9.0	-	-	+7000	-	-	RNP1
002	TF	BIJIN	-	039 (030.3)	-9.0	15.9	-	+4200	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	CHOKA	003 (353.8)	-9.0	1.0(-14000)	R	8000	FL140	-230(-14000)	RNP1

STANDARD ARRIVAL CHART-INSTRUMENT

RJSK / AKITA

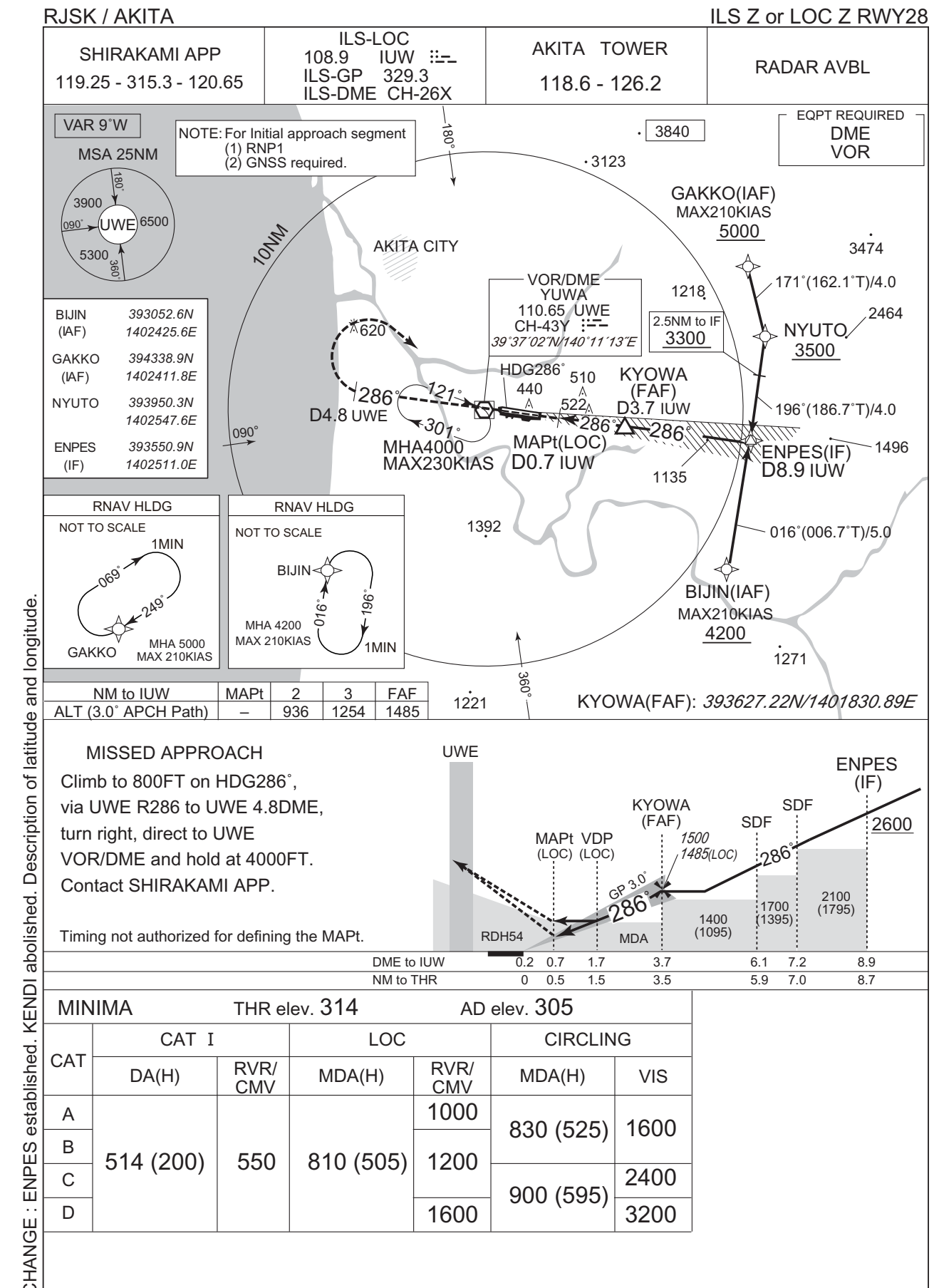
RNAV STAR RWY28

Waypoint Coordinates

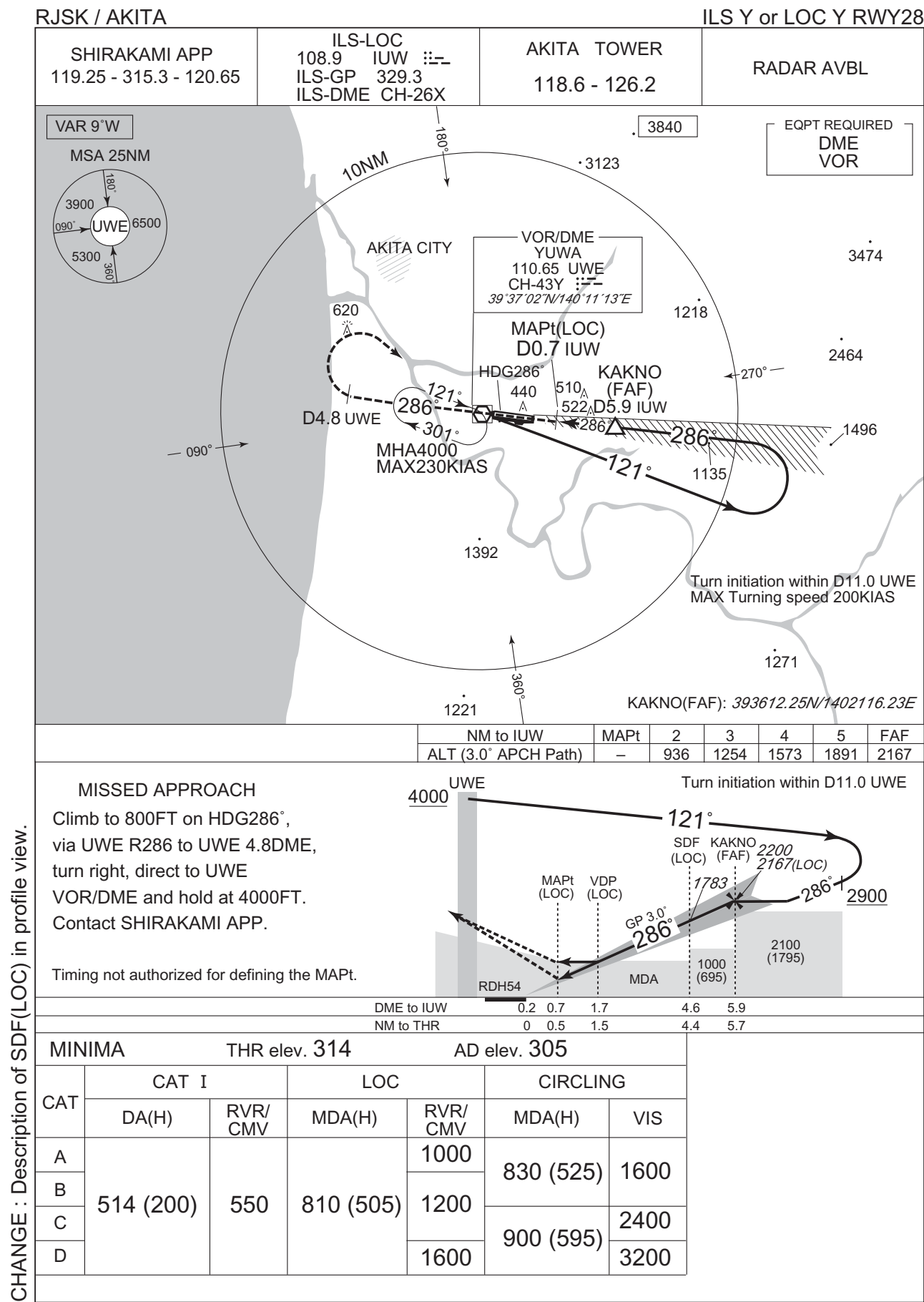
Waypoint Identifier	Coordinates
MAGGY	395456.6N / 1401926.8E
GAKKO	394338.9N / 1402411.8E
YAYOI	391910.1N / 1395933.8E
CHOKA	391709.5N / 1401401.1E
BIJIN	393052.6N / 1402425.6E

CHANGE : Waypoint Coordinates added.

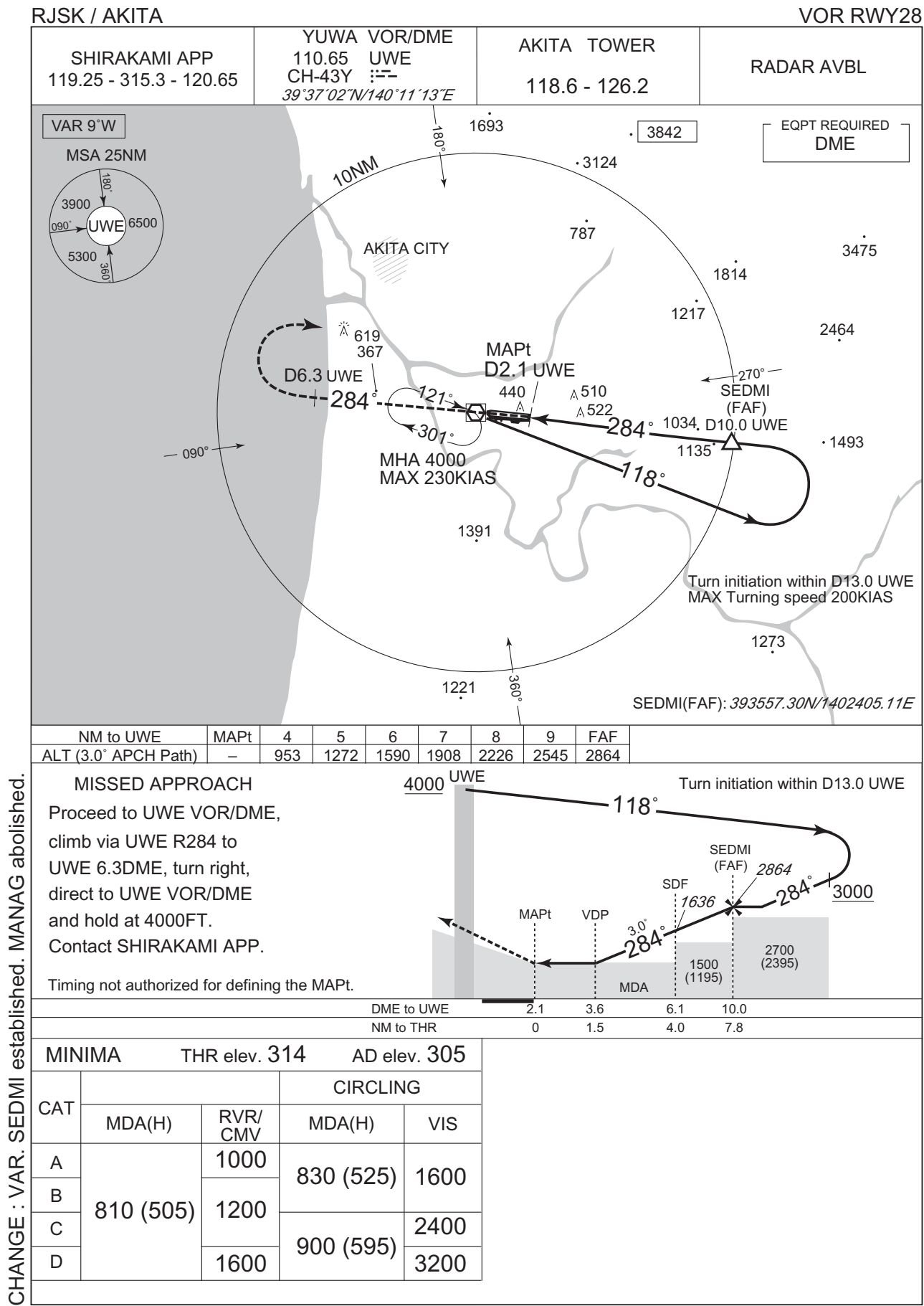
INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART



RJSK / AKITA

SHIRAKAMI APP
119.25 - 315.3 - 120.65

YUWA VOR/DME
110.65 UWE
CH-43Y
39°37'02"N/140°11'13"E

AKITA TOWER
118.6 - 126.2

RADAR AVBL

VAR 9°W

EQPT REQUIRED
DME

YAESU (FAF) D5.0 UWE

MAPt 440

MHA 4000 MAX 230KIAS

Turn initiation within D11.0 UWE
MAX Turning speed 200KIAS

NM to UWE	FAF	4	3	2	1	MAPt
ALT (3.0° APCH Path)	2185	1866	1548	1229	911	—

MISSED APPROACH

Climb via UWE R100 to UWE 8.6DME, turn right, direct to UWE VOR/DME and hold at 4000FT.

Contact SHIRAKAMI APP.

Timing not authorized for defining the MAPt.

10.8		5.0	3.0	0.1	DME to UWE
11.6		5.8	3.8	0.9	NM to THR

MINIMA

THR elev. 289

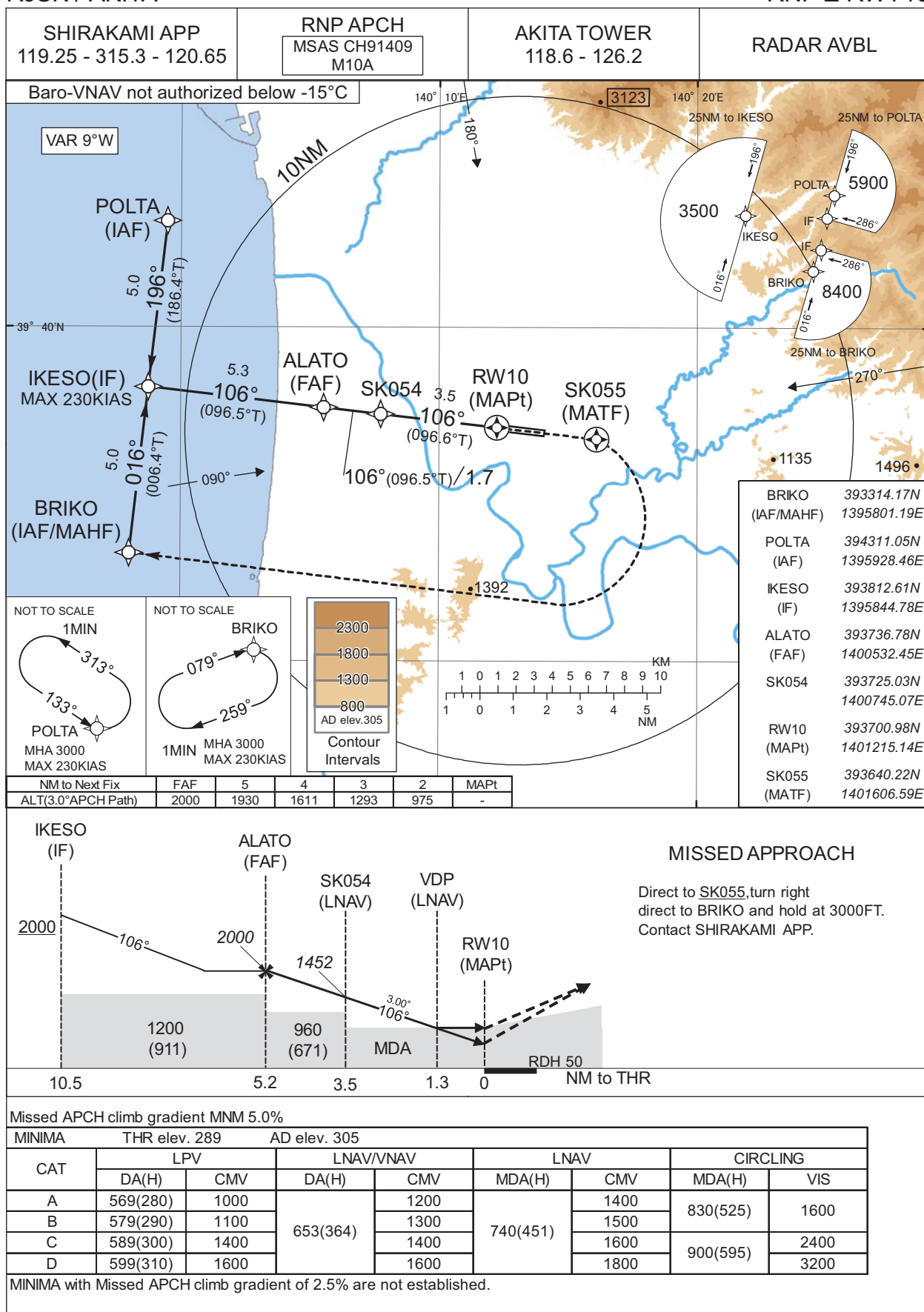
AD elev. 305

CAT			CIRCLING	
	MDA(H)	CMV	MDA(H)	VIS
A	680 (391)	1200	830 (525)	1600
B		1300		
C		1400	900 (595)	2400
D		1600		

CHANGE : PROC renamed. VAR.

RJSK / AKITA

RNP Z RWY10



CHANGE : IKESO, ALATO established. KILLY, TAMPO abolished.

INSTRUMENT APPROACH CHART

RJSK / AKITA

RNP Z RWY10

FAS DATA BLOCK

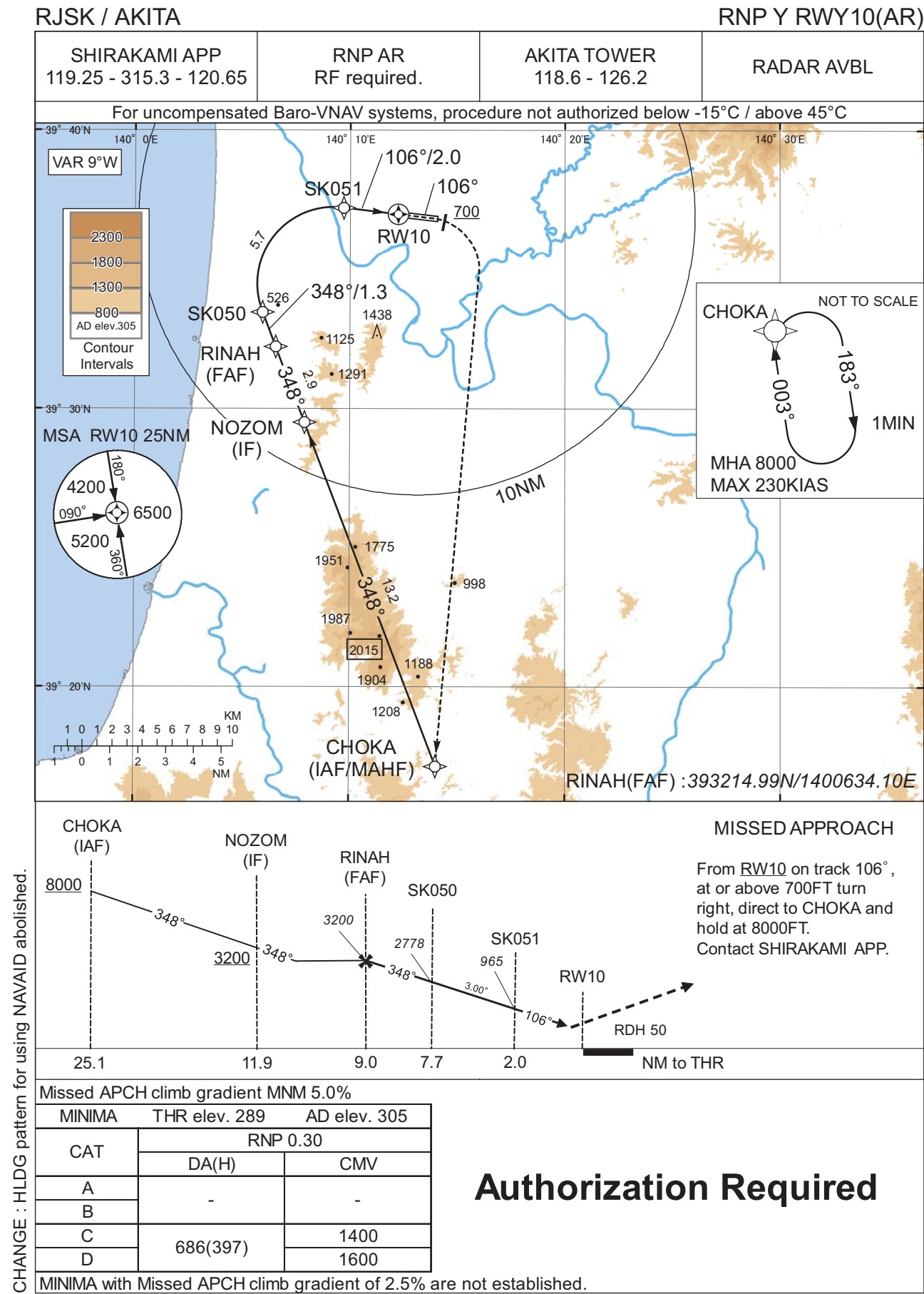
Operation type	0	LTP/FTP ellipsoidal height	+01268
SBAS service provider identifier	2	FPAP latitude	393651.6070N
Airport identifier	RJSK	FPAP longitude	1401359.3150E
Runway	10	Threshold crossing height	00015.0
Approach performance designator	0	TCH units selector	1
Route indicator	Z	Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M10A	∠ length offset	0000
LTP/FTP latitude	393700.9295N	HAL	40.0
LTP/FTP longitude	1401215.1950E	VAL	50.0
CRC remainder	884983FD		

Required additional data

LTP/FTP orthometric height	87.9
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CHANGE : FAS DATA BLOCK; Required additional data established.

INSTRUMENT APPROACH CHART



CHOKA (IAF)

NOZOM (IF)

RINAH (FAF)

SK050

SK051

RW10

RDH 50

NM to THR

MISSED APPROACH

From RW10 on track 106°, at or above 700FT turn right, direct to CHOKA and hold at 8000FT. Contact SHIRAKAMI APP.

Missed APCH climb gradient MNM 5.0%

MINIMA	THR elev. 289	AD elev. 305
CAT	RNP 0.30	
	DA(H)	CMV
A	-	-
B	-	-
C	686(397)	1400
D	686(397)	1600

MINIMA with Missed APCH climb gradient of 2.5% are not established.

Authorization Required

INSTRUMENT APPROACH CHART

RJSK / AKITA

RNP Y RWY10(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	CHOKA	-	-	-9.0	-	-	+8000	-	-	-
002	TF	NOZOM	-	348 (339.2)	-9.0	13.2	-	+3200	-	-	1.0
003	TF	RINAH	-	348 (339.1)	-9.0	2.9	-	3200	-	-	1.0
004	TF	SK050	-	348 (339.1)	-9.0	1.3	-	2778	-	-3.00	0.3
005	RF Center: SKRF1 r=2.78NM	SK051	-	-	-9.0	5.7	R	965	-	-3.00	0.3
006	TF	RW10	Y	106 (096.6)	-9.0	2.0	-	339	-	-3.00/50	0.3
007	FA	-	-	106 (096.6)	-9.0	-	-	+700	-	-	1.0
008	DF	CHOKA	-	-	-9.0	-	R	8000	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	CHOKA	003 (353.8)	-9.0	1.0 (-14000)	R	8000	FL140	-230 (-14000)	1.0

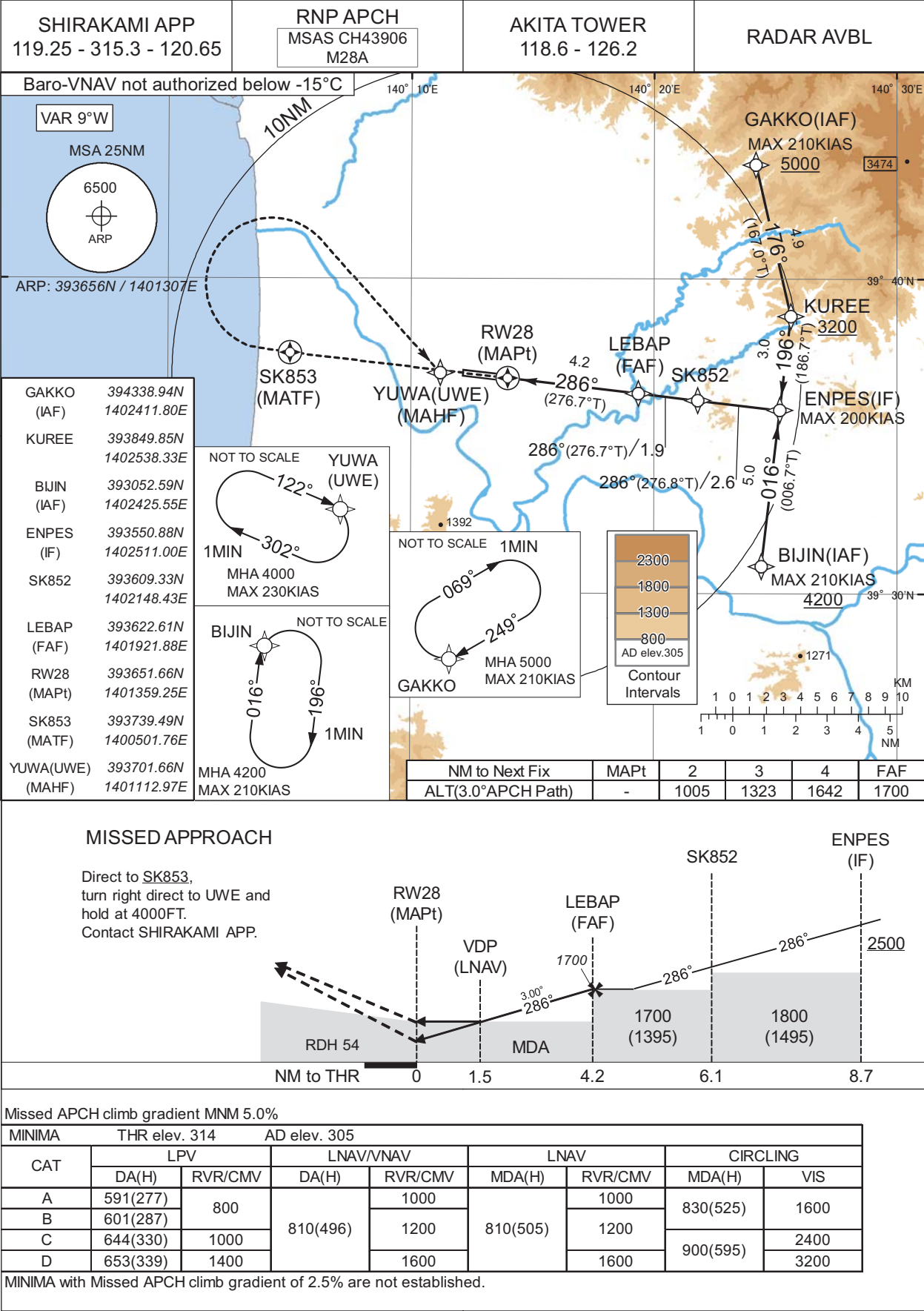
Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
CHOKA	391709.51N / 1401401.06E	SKRF1	393428.91N / 1400918.43E
NOZOM	392931.83N / 1400754.90E		
RINAH	393214.99N / 1400634.10E		
SK050	393329.21N / 1400557.30E		
SK051	393714.55N / 1400943.07E		
RW10	393700.98N / 1401215.14E		

INSTRUMENT APPROACH CHART

RJSK / AKITA

RNP Z RWY28



INSTRUMENT APPROACH CHART

RJSK / AKITA

RNP Z RWY28

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+01346
SBAS service provider identifier	2	FPAP latitude	393700.9295N
Airport identifier	RJSK	FPAP longitude	1401215.1950E
Runway	28	Threshold crossing height	00016.5
Approach performance designator	0	TCH units selector	1
Route indicator	Z	Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M28A	∟ length offset	0000
LTP/FTP latitude	393651.6070N	HAL	40.0
LTP/FTP longitude	1401359.3150E	VAL	50.0
CRC remainder	CA86C5B4		

Required additional data

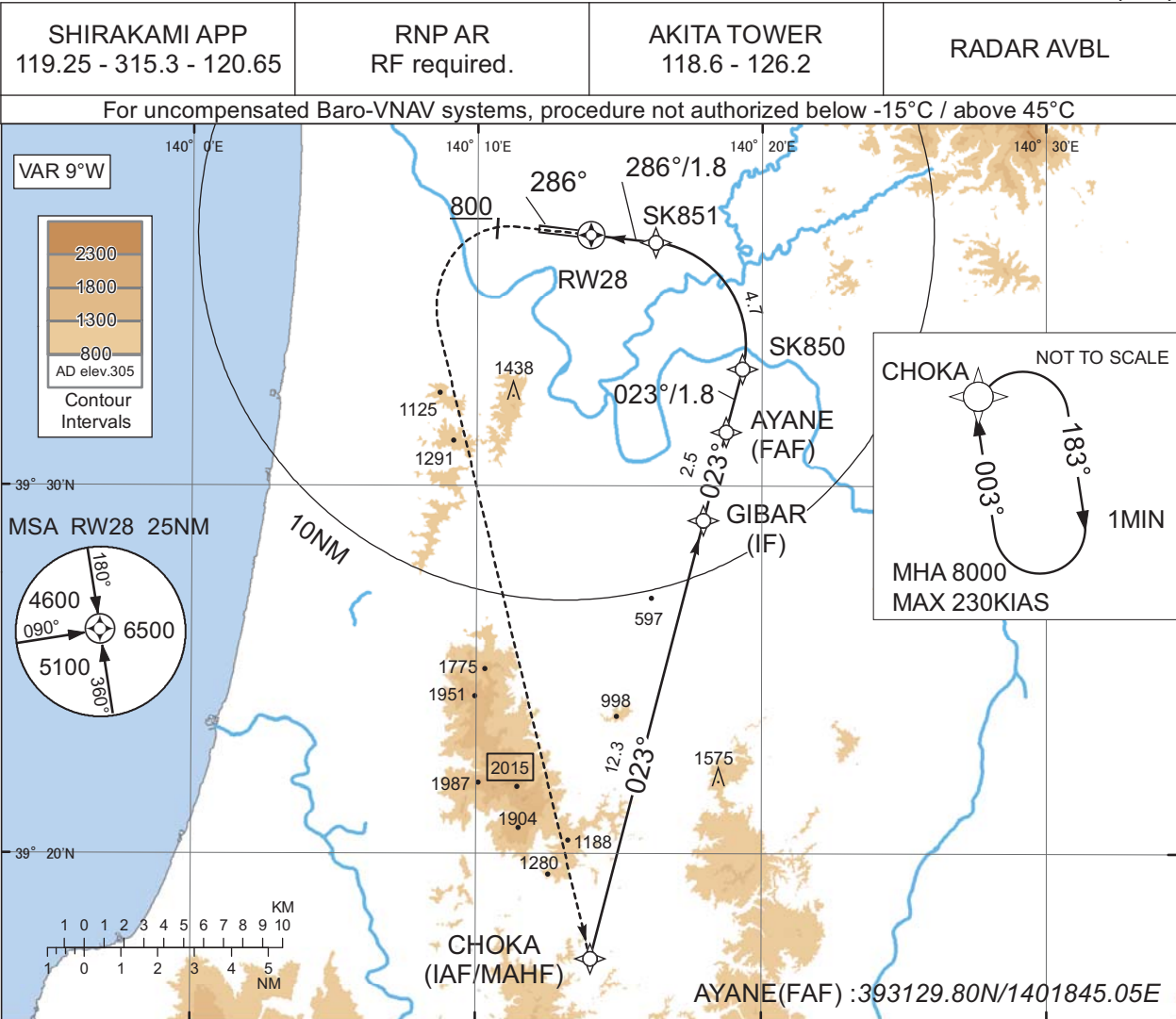
LTP/FTP orthometric height	95.6
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CHANGE : New PROC.

INSTRUMENT APPROACH CHART

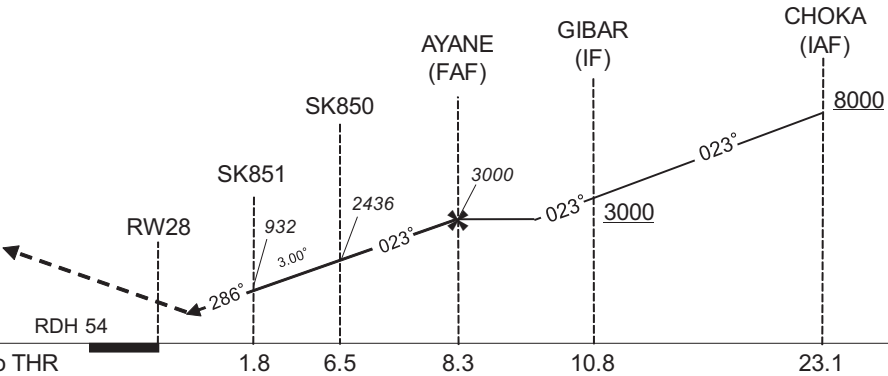
RJSK / AKITA

RNP Y RWY28(AR)



MISSED APPROACH

From RW28 on track 286°, at or above 800FT turn left, direct to CHOKA and hold at 8000FT. Contact SHIRAKAMI APP.



Missed APCH climb gradient MNM 5.0%

MINIMA	THR elev. 314	AD elev. 305
CAT	RNP 0.30	
	DA(H)	RVR/CMV
A	-	-
B	-	-
C	653(339)	1000
D		1400

Authorization Required

MINIMA with Missed APCH climb gradient of 2.5% are not established.

CHANGE : PROC renamed. HLDG pattern for using NAVAID abolished.

INSTRUMENT APPROACH CHART

RJSK / AKITA

RNP Y RWY28(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	CHOKA	-	-	-9.0	-	-	+8000	-	-	-
002	TF	GIBAR	-	023 (014.3)	-9.0	12.3	-	+3000	-	-	1.0
003	TF	AYANE	-	023 (014.3)	-9.0	2.5	-	3000	-	-	1.0
004	TF	SK850	-	023 (014.3)	-9.0	1.8	-	2436	-	-3.00	0.3
005	RF Center: SKRF2 r=2.77NM	SK851	-	-	-9.0	4.7	L	932	-	-3.00	0.3
006	TF	RW28	Y	286 (276.7)	-9.0	1.8	-	368	-	-3.00/54	0.3
007	FA	-	-	286 (276.7)	-9.0	-	-	+800	-	-	1.0
008	DF	CHOKA	-	-	-9.0	-	L	8000	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	CHOKA	003 (353.8)	-9.0	1.0 (-14000)	R	8000	FL140	-230 (-14000)	1.0

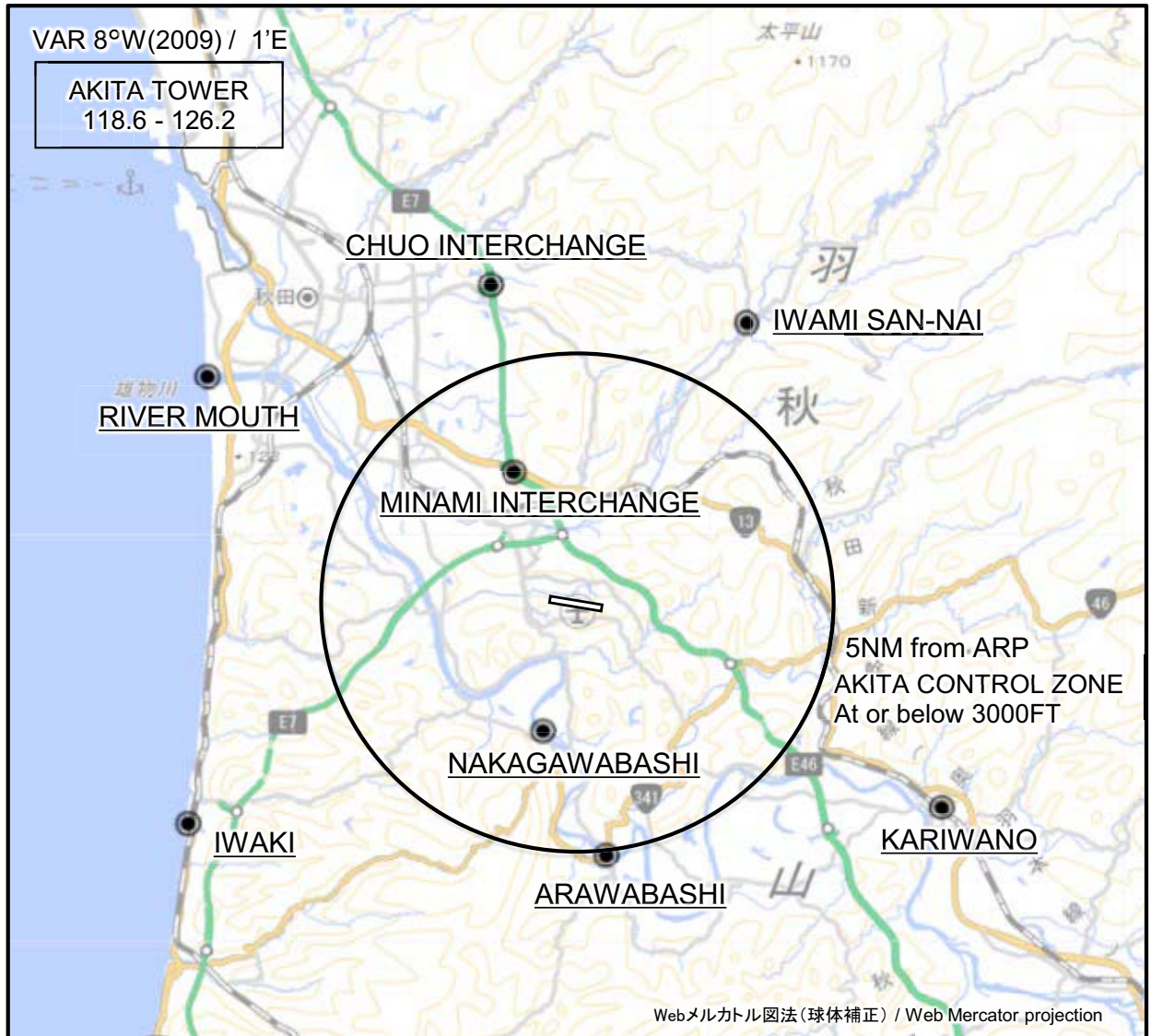
Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
CHOKA	391709.51N / 1401401.06E	SKRF2	393354.17N / 1401551.05E
GIBAR	392905.47N / 1401757.27E		
AYANE	393129.80N / 1401845.05E		
SK850	393312.87N / 1401919.21E		
SK851	393639.39N / 1401615.88E		
RW28	393651.66N / 1401359.25E		

CHANGE : PROC renamed.

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Visual REP



※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

CHANGE : Map updated. BRG/DIST from ARP.

Call sign	BRG / DIST from ARP	Remarks
中央インターチェンジ Chuo Interchange	345°T / 6.6NM	秋田中央インターチェンジ(秋田自動車道) Akita Chuo Interchange
岩見三内 Iwami San-nai	030°T / 6.5NM	岩見川と三内川の合流点 Merging point of Iwamigawa River and San-naigawa River
リバーマウス River Mouth	302°T / 8.6NM	雄物川河口 Omonogawa River Mouth
南インターチェンジ Minami Interchange	334°T / 3.0NM	秋田南インターチェンジ(秋田自動車道) Akita Minami Interchange
中川橋 Nakagawabashi	196°T / 2.6NM	中川橋(雄物川) Nakagawabashi
刈和野 Kariwano	120°T / 8.2NM	JR刈和野駅 JR Kariwano Station
岩城 Iwaki	240°T / 8.8NM	道の駅岩城 Michinoeki (Road Station) Iwaki
新波橋 Arawabashi	174°T / 5.0NM	新波橋(雄物川) Arawabashi

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Minimum Vectoring Altitude CHART

